Central Tendency

September 19, 2022

Review: Levels of Measurement

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Ordinal variables are those whose values can be rank ordered but do not have equal unit differences.

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We often want to summarize data, but there are different ways to doing so depending on the level of measurement.

Penn State Football Dataset

Opponent	PSU Score	Opponent Score	PSU Win/Loss	PSU Ranking
Wisconsin	16	10	Win	19
Ball State	44	13	Win	11
Auburn	28	20	Win	10
Villanova	38	17	Win	6
Indiana	24	0	Win	4
Iowa	20	23	Loss	4
Illinois	18	20	Loss	7
Ohio State	24	33	Loss	20
Maryland	31	14	Win	22
Michigan	17	21	Loss	23
Rutgers	28	0	Win	Not ranked
Michigan State	27	30	Loss	Not ranked
Arkansas	10	24	Loss	Not ranked
Purdue	35	31	Win	Not ranked
Ohio	46	10	Win	Not ranked

Frequency Distribution (Intro)

An overview of <u>all</u> distinct values in a variable and the number of times they are observed.

Outcomes	**Frequency**
Wins	9
Losses	6
Total Games =	15

Frequency Distribution (Intro)

An overview of <u>all</u> distinct values in a variable and the number of times they are observed.

- More on frequency distributions in the next week or so...

Points	**Frequency**	**Points x Frequency**
10	1	10x1 = 10
16	1	16x1 = 16
17	1	17x1=17
18	1	18x1=18
20	1	20x1=20
24	2	24x2=48
27	1	27x1=27
28	2	28x2=56
31	1	31x1=31
35	1	35x1=35
38	1	38x1=38
44	1	44x1=44
46	1	46x1=46
	Total Points =	**406**

Measures of Central Tendency

A single value representing the most typical, common, or representative value in a sample or population.

- Mean
- Median
- Mode

Interlude: Statistical Notation and Terms

Parameter: summary value for a *population*

Statistic: summary value for a *sample*

Outlier: An excessively large or excessively small value relative to the rest of the data

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Notation:

- bars above variables indicate "mean" (e.g., $\bar{X}$)
- subscripts index data (e.g., $\bar{X}_1$ means the first observation in variable X, $\bar{X}_2$ means the second observation, and $\bar{X}_i$ represents "values of X" generally.)
- $\sum_{i=1}^{n}$ means "summation" (e.g., $\sum_{i=1}^{n} X_i$ means add up values for all observations in X)
  - top number is ending point of summation- n usually denotes "sample size"
  - bottom number is starting point of summation

(Arithmetic) Mean

The sum of all observations divided by the sample size

$$\bar{X} = \sum_{i=1}^{n} \frac{X_i}{n} = \frac{1}{n} \sum_{i=1}^{n} X_i \tag{1}$$

Good for numeric, continuous data

- Never use for categorical data!
- Very sensitive to outliers!

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Calculating the mean

Mean PSU scores: Total Points/Total Games

$$\frac{(16+44+28+38+24+20+18+24+31+17+28+27+10+35+46)}{15} = 27.07$$

Calculating the mean

Mean PSU scores: Total Points/Total Games

$$\frac{(16+44+28+38+24+20+18+24+31+17+28+27+10+35+46)}{15} = 27.07$$

Mean Opponent scores: Total Points/Total Games

$$\frac{(10+13+20+17+0+23+20+33+14+21+0+30+24+31+10)}{15} = 17.73$$

Median

The middle value of ordered data. The 50th percentile.

1. Rank order the data from lowest to highest
2. Find the value for the observation corresponding to $\frac{n+1}{2}$
3. If n is even, take the mean of the two middling values

Best for ordinal data (usually), but also
<u>not sensitive to outliers in continuous interval/ratio data!</u>

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Calculating the Median

Median PSU Points per game: $\frac{15+1}{2} = 8$th observation...

[10, 16, 17, 18, 20, 24, 24, **27**, 28, 28, 31, 35, 38, 44, 46]

Calculating the Median

Median PSU Points per game: $\frac{15+1}{2} = 8$th observation...

[10, 16, 17, 18, 20, 24, 24, **27**, 28, 28, 31, 35, 38, 44, 46]

Median PSU Ranking (from ranked weeks only): $\frac{10+1}{2} = 5.5$th observation...

[4, 4, 6, 7, **10**, **11**, 19, 20, 22, 23] $=$ **10.5**

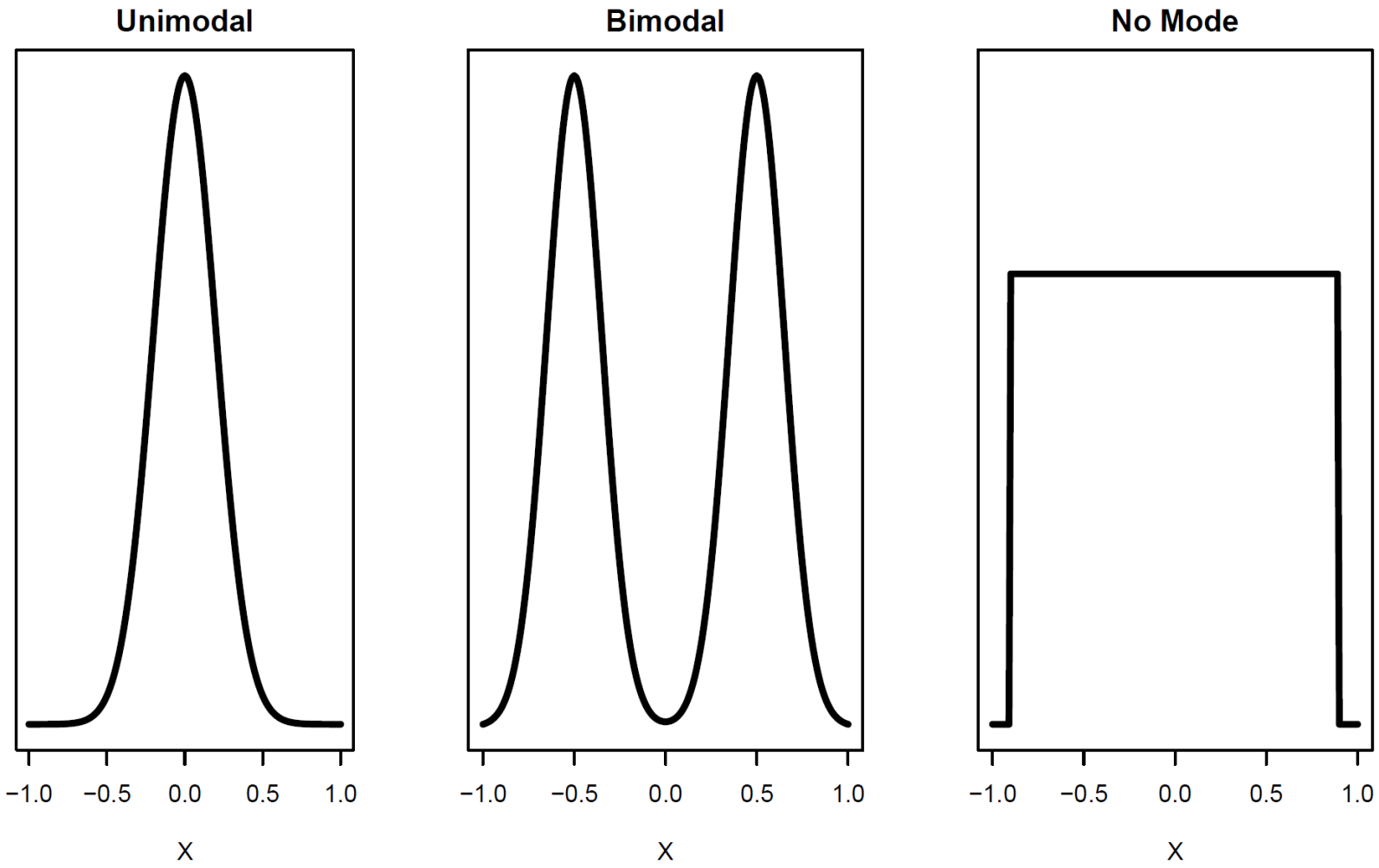
Mode

The most common observation (observation with the highest frequency)

Characteristics:

- Usually best for categorical data, but technically can be used at all levels
- The modal value isn't necessarily "central"
- A distribution can have multiple modes
- A distribution can have no mode

Mode



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Calculating Modes

Modal PSU score: 24 and 28 (both happen twice)

Modal game outcome: **Win** (9 out of 15)

Modal Ranking: **Not ranked**

- Or, among only ranked weeks: **4** (twice)

To R! Goals:

1. Create a Folder and Rscript for your projects
2. Create a new Rscript for today
3. Set working directory
4. Load in data
5. View dataset
6. View variable in dataset
7. Calculate measures of central tendency